order40 RESULTS: RG37, (0,1)-edge decomposition

PREREG sha256 069e6712e02f349c9cb8a27d445ec7aa30738c631216e8c00f41d972d647f031 (hashed before numerics). Stored arrays only (order19 C2a/C2b, order34 pieces).

I-e2: T_e2((1,n),(0,1)) = ((1+n)/2)·C2a, n = 1..6: PASS

Max error 1.2e−12. This is the e2 analogue of T_e1 = ((2n+7)/4)·C2b. Q(e2) = 7/2 enters only the coefficient B/(4Q(e2)) = (7+7n)/14. It is Category A only in the bookkeeping sense (immediate from the definitions).

RG37a: PASS (consistency)

r(T_e2) = r(C2a) = 0.020718 on 6/6 cones (identical to < 1e−12), and the criterion is met. The near-law carrier on (0,1)-edge cones is T_e2, a multiple of the single Bernoulli sum C2a.

RG37b: PASS (Category B; not blind; order34 observed these values)

cone	r(T_e2)	r(B2 part)	r(T_BB)
((1,1),(0,1))	0.02072	0.02317	0.265
((1,2),(0,1))	0.02072	0.02276	0.397
((1,3),(0,1))	0.02072	0.02225	0.378
((1,4),(0,1))	0.02072	0.02257	0.439
((1,5),(0,1))	0.02072	0.02167	0.432
((1,6),(0,1))	0.02072	0.02196	0.524

The B2 part sits within [½, 2]·r(C2a) on 6/6 cones, and T_BB fails on 6/6.

Controls

- r(C2b) = 0.0030492, matching the (1,0)-edge value (order34).
- Interior cones: 0/44 B2 pieces are a scalar multiple of C2a or C2b (best-scale residual ≥ 1e−3), PASS.

Reading

The edge rule is now explicit:

- an e1 edge gives a B2 piece ∝ C2b, with r = 0.00305;
- an e2 edge gives a B2 piece ∝ C2a, with r = 0.0207;
- an interior cone gives B2 pieces that are sheared sums, worse (r 0.04–0.17).

The near-law grade is the law miss of the coordinate Bernoulli sum attached to the edge. Q(e) only scales.

Category

A: I-e2 (bookkeeping identity, too trivial for the paper). B: RG37b. v11: no. Ξ: not identified. E1–E3 OPEN.

order40 NOTES

- The two coordinate Bernoulli sums C2b (ρ2) and C2a (ρ1) have law misses 0.00305 and 0.0207. Why C2a is 7× worse is open; a natural suspect is the asymmetry Q(e1) = 1 vs Q(e2) = 7/2 in L_B-type aliasing. L_B is proved only for ρ2.
- FORWARD notes written in order30/ and order34/ (new files only).
- Next: order41 RG38 (two-variable design), order42 RG39 (L_B at integer x).
- Naming: RG* only, no labels L1–L7; "36" only as dim V. MAPS_CANDIDATES and REVIEW untouched; no Resonant objects; no new Bernoulli objects; no bridge claim; CORE_FREEZE. nice, OMP_NUM_THREADS=1; no push or sends; new files only.